

Homework #5 (Submit before 19-May)

❖ Ch3_No.5

For the two-energy levels system, the spontaneous emission lifetime of energy level E_2 is τ_{s2} , and the nonradiative recombination lifetime is τ_{nr2} . If the carrier density at energy level E_2 at $t=0$ is $n_2(0)$, the volume of the gain material is V , the frequency of the spontaneous emission is ν , then please calculate:

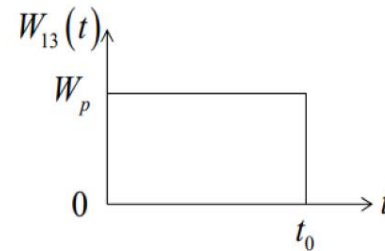
- 1) The optical power of the spontaneous emission change with time t .
- 2) The photon number of the spontaneous emission from the carriers in energy level E_2 during the decay process.
- 3) The ratio η_2 of the spontaneous emission photon number with the initial carrier number at energy level E_2 . η_2 is known as the quantum yield.

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❖ Ch3_No.8

Assuming that the ruby with a carrier density of n is excited by a rectangular pulse, the pumping transition probability can be expressed as

$$W_{13}(t) = \begin{cases} W_p & 0 < t \leq t_0 \\ 0 & t > t_0 \end{cases}$$



Please determine the upper carrier density $n_2(t)$, and then draw the corresponding curve.

(Note: S_{32} is very big and therefore $n_3 \approx 0$, $dn_3/dt = 0$. The stimulated transition between the up and down energy levels can be ignored without optical cavity.)